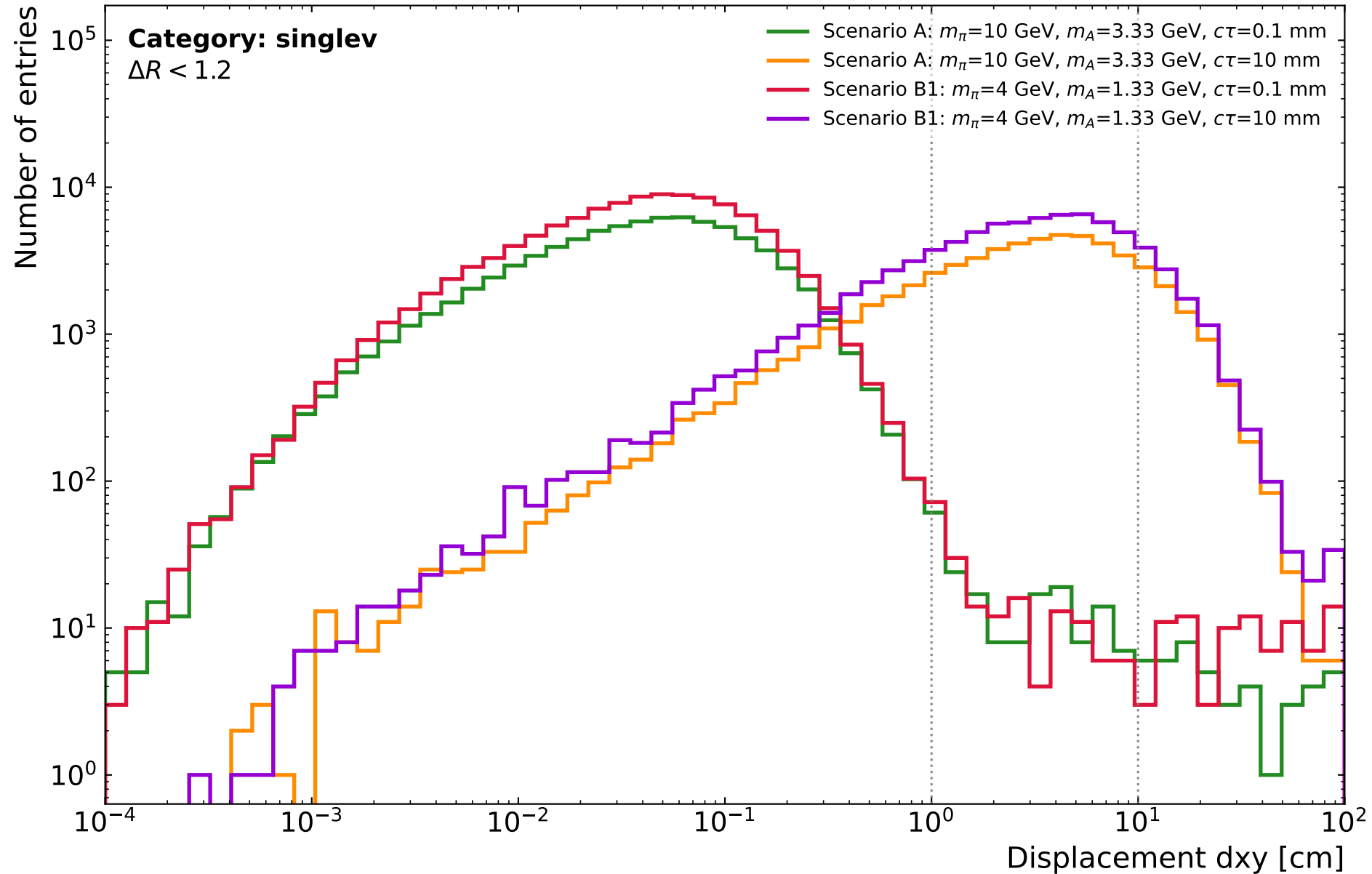


singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

CMS *Simulation Preliminary*

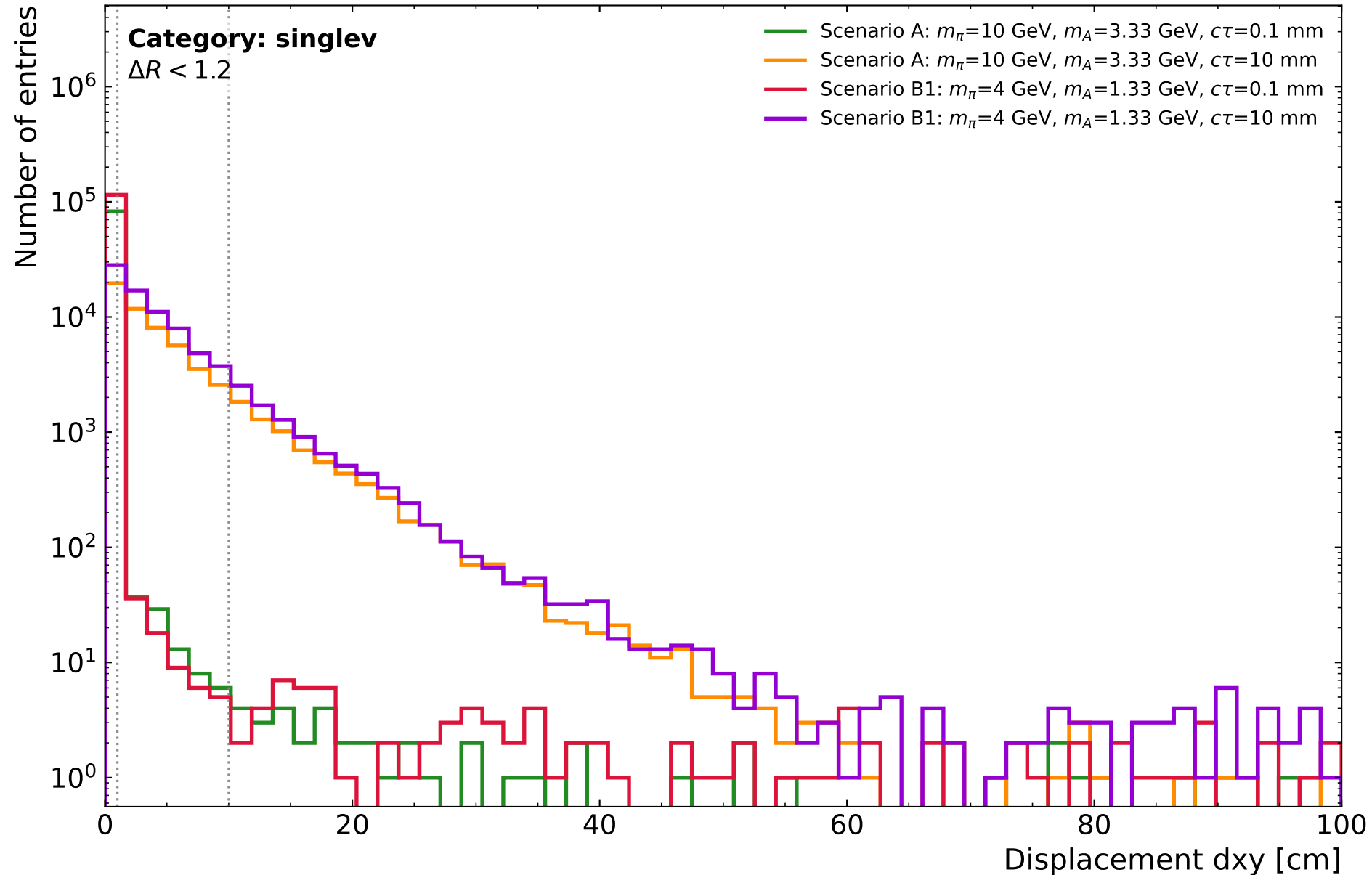
(13.6 TeV, 2024)



singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

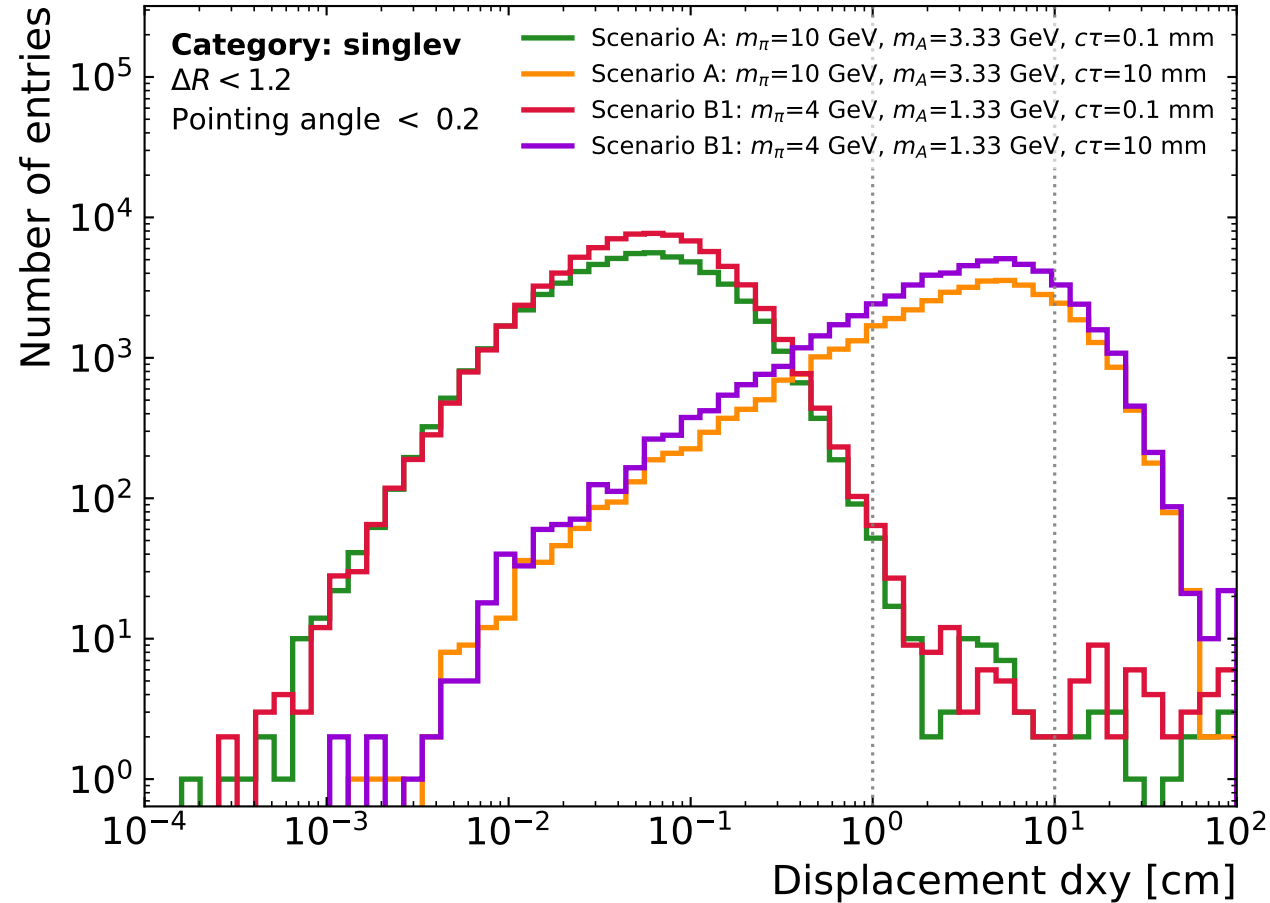
CMS *Simulation Preliminary*

(13.6 TeV, 2024)



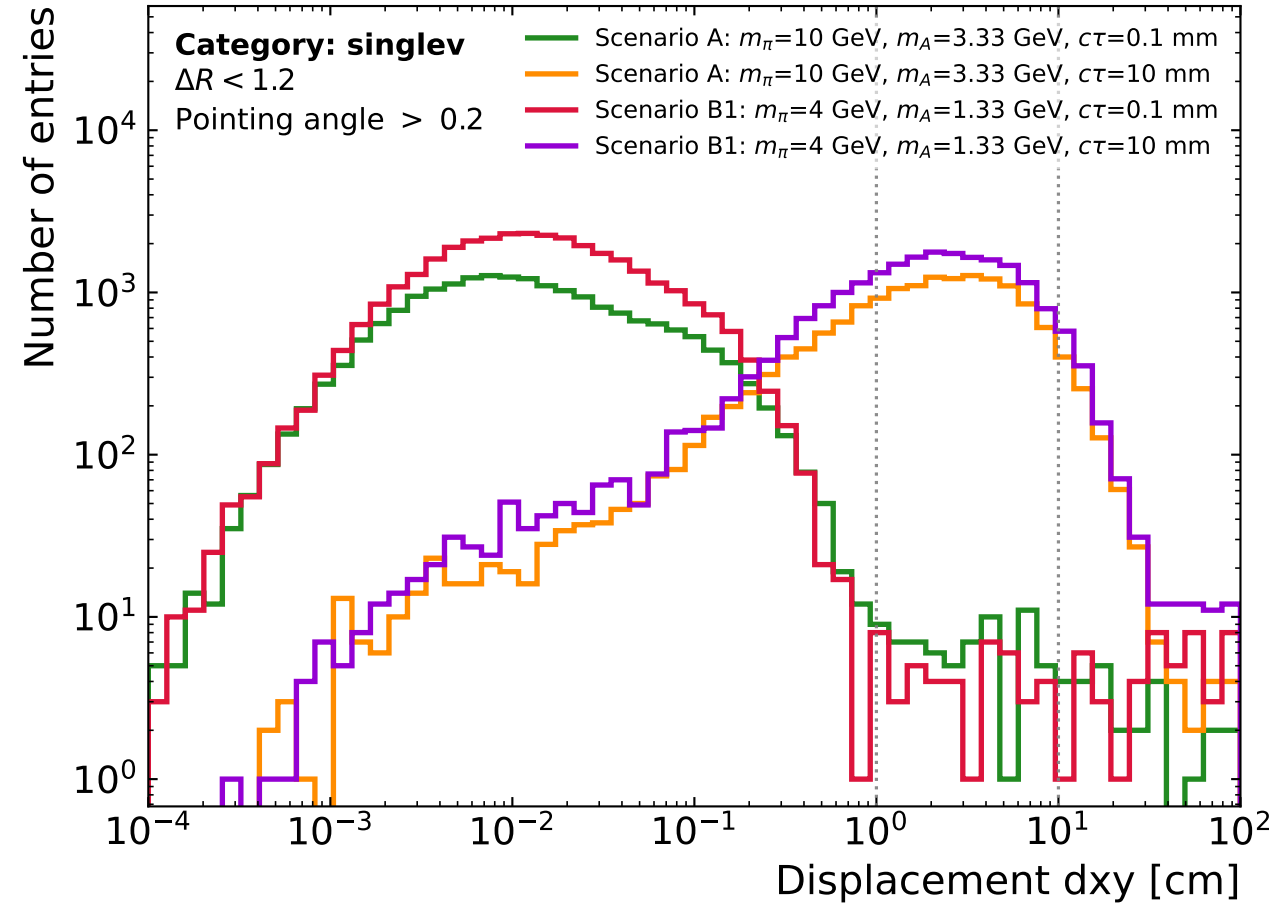
singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	62,621	89	18	62,728
scA $c\tau=10$	8,231	27,603	6,701	42,535
scB1 $c\tau=0.1$	81,039	109	47	81,195
scB1 $c\tau=10$	11,943	39,543	8,557	60,043

CMS Simulation Preliminary (13.6 TeV, 2024)

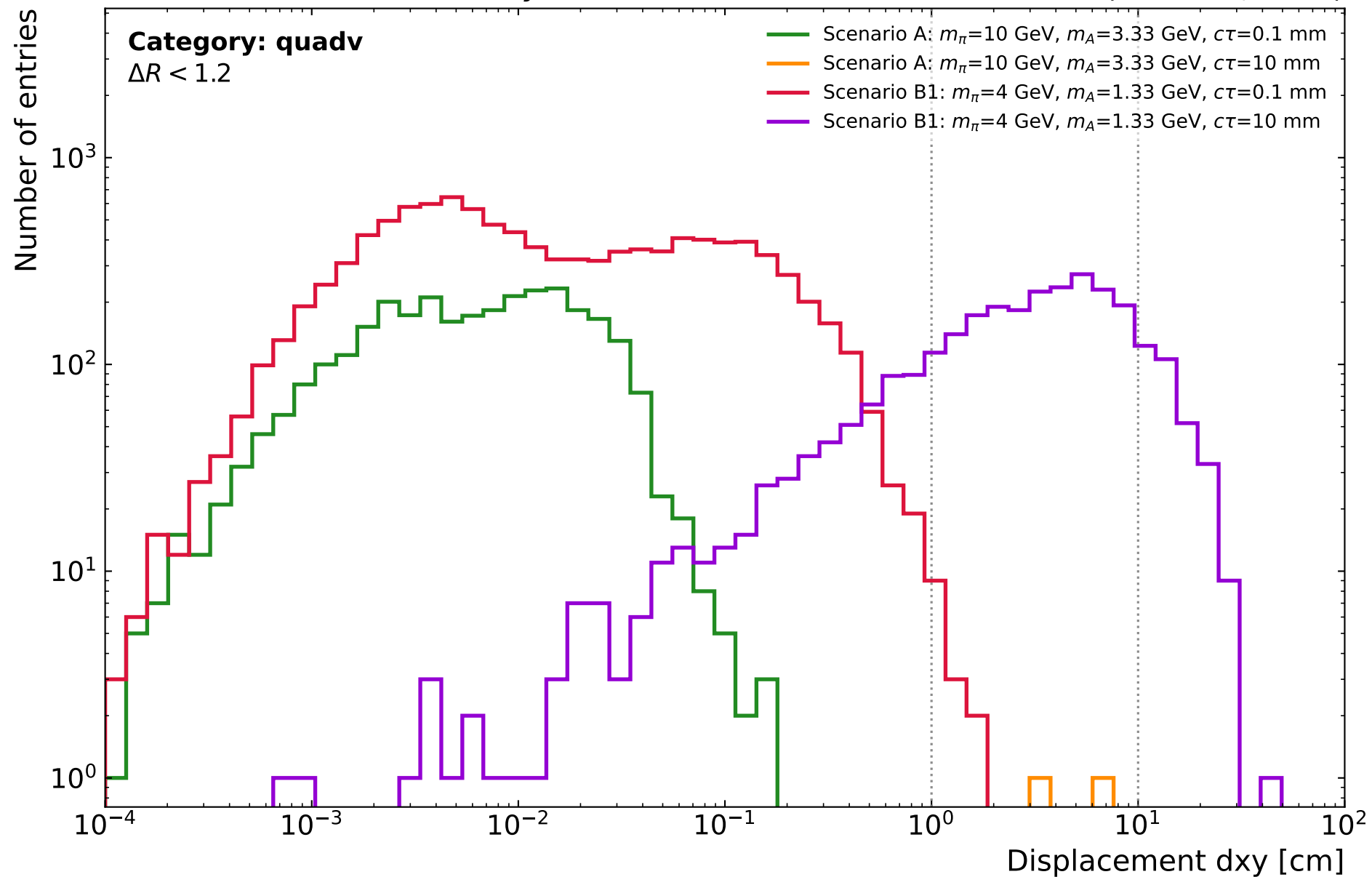


	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	19,811	65	27	19,903
scA $c\tau=10$	4,774	10,367	812	15,953
scB1 $c\tau=0.1$	33,815	42	48	33,905
scB1 $c\tau=10$	6,601	14,313	1,138	22,052

quadv: displacement of the binning vertex (fourmuonSV_dxy)
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

CMS *Simulation Preliminary*

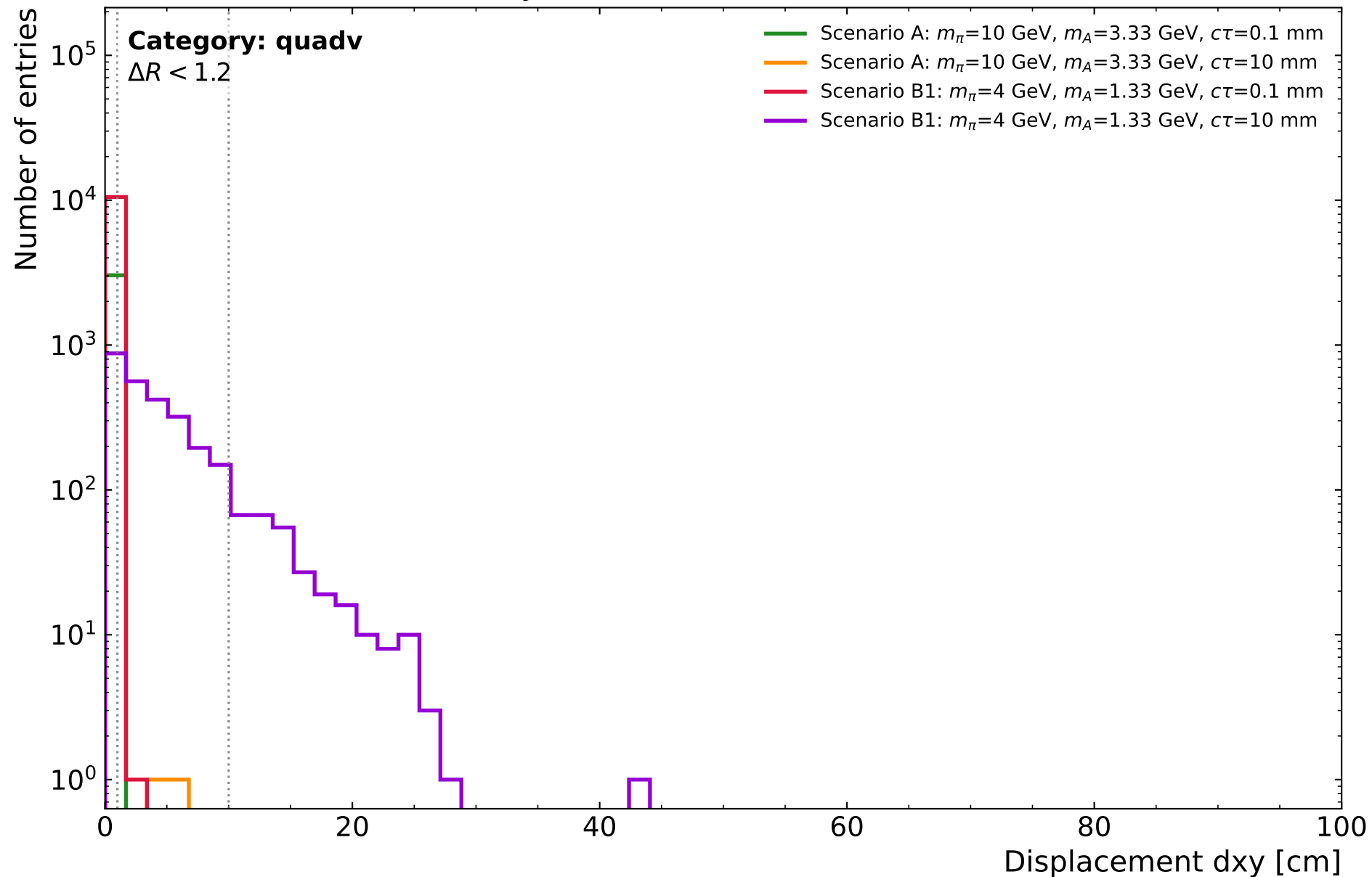
(13.6 TeV, 2024)



quadv: displacement of the binning vertex (fourmuonSV_dxy)
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

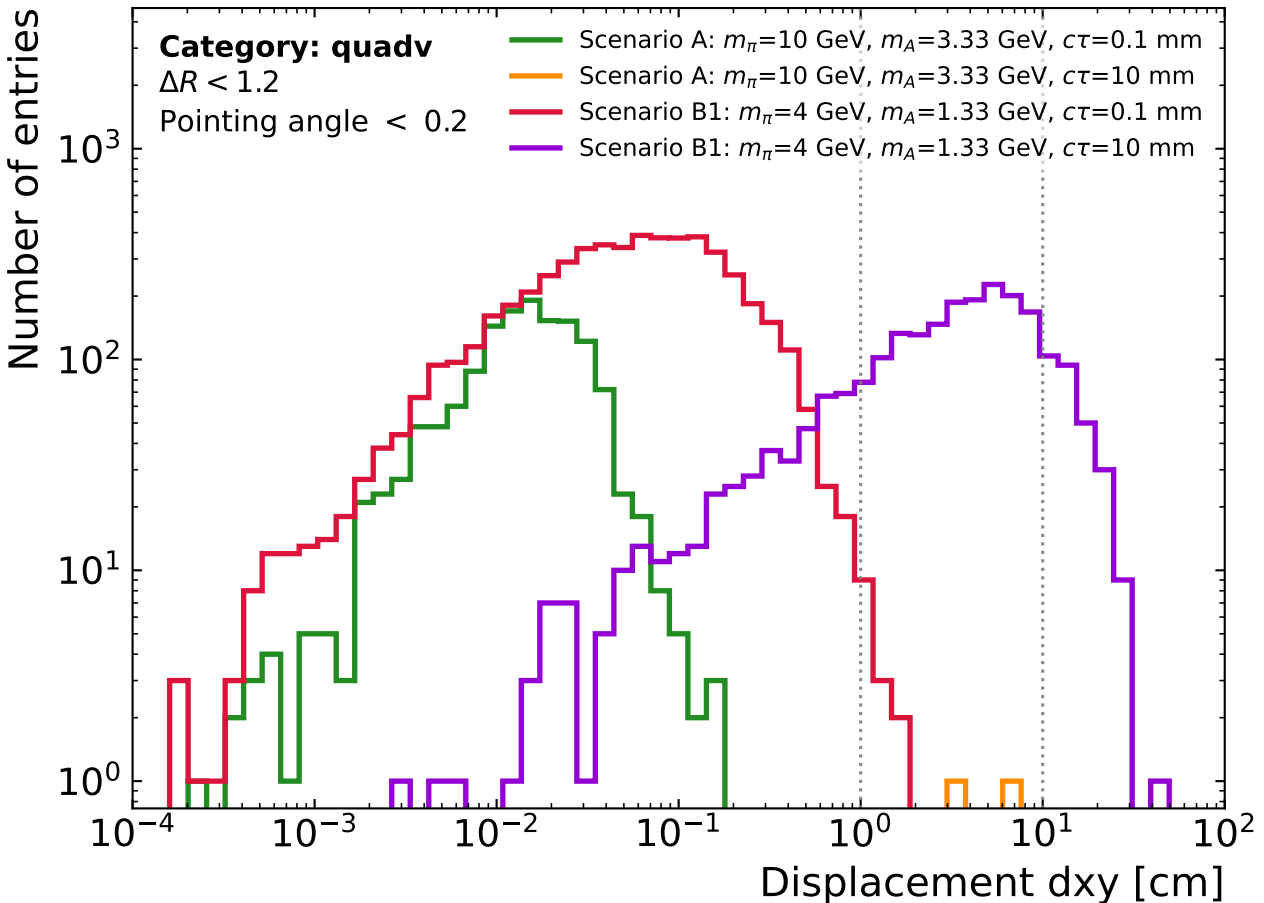
CMS *Simulation Preliminary*

(13.6 TeV, 2024)



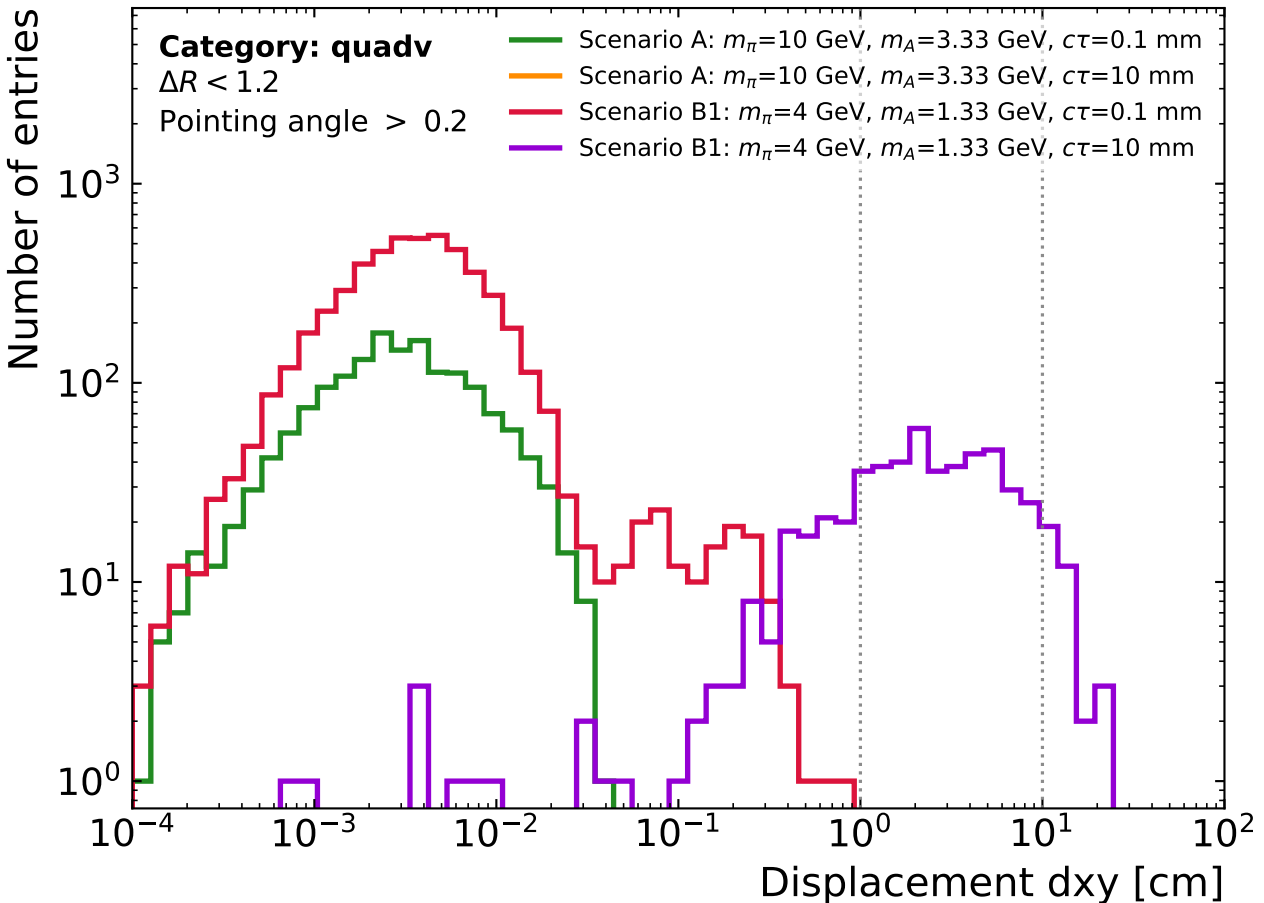
quadv: displacement of the binning vertex (fourmuonSV_dxy), split by pointing angle
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	1,402	0	0	1,402
scA $c\tau=10$	0	2	0	2
scB1 $c\tau=0.1$	5,332	11	0	5,343
scB1 $c\tau=10$	449	1,553	267	2,269

CMS Simulation Preliminary (13.6 TeV, 2024)

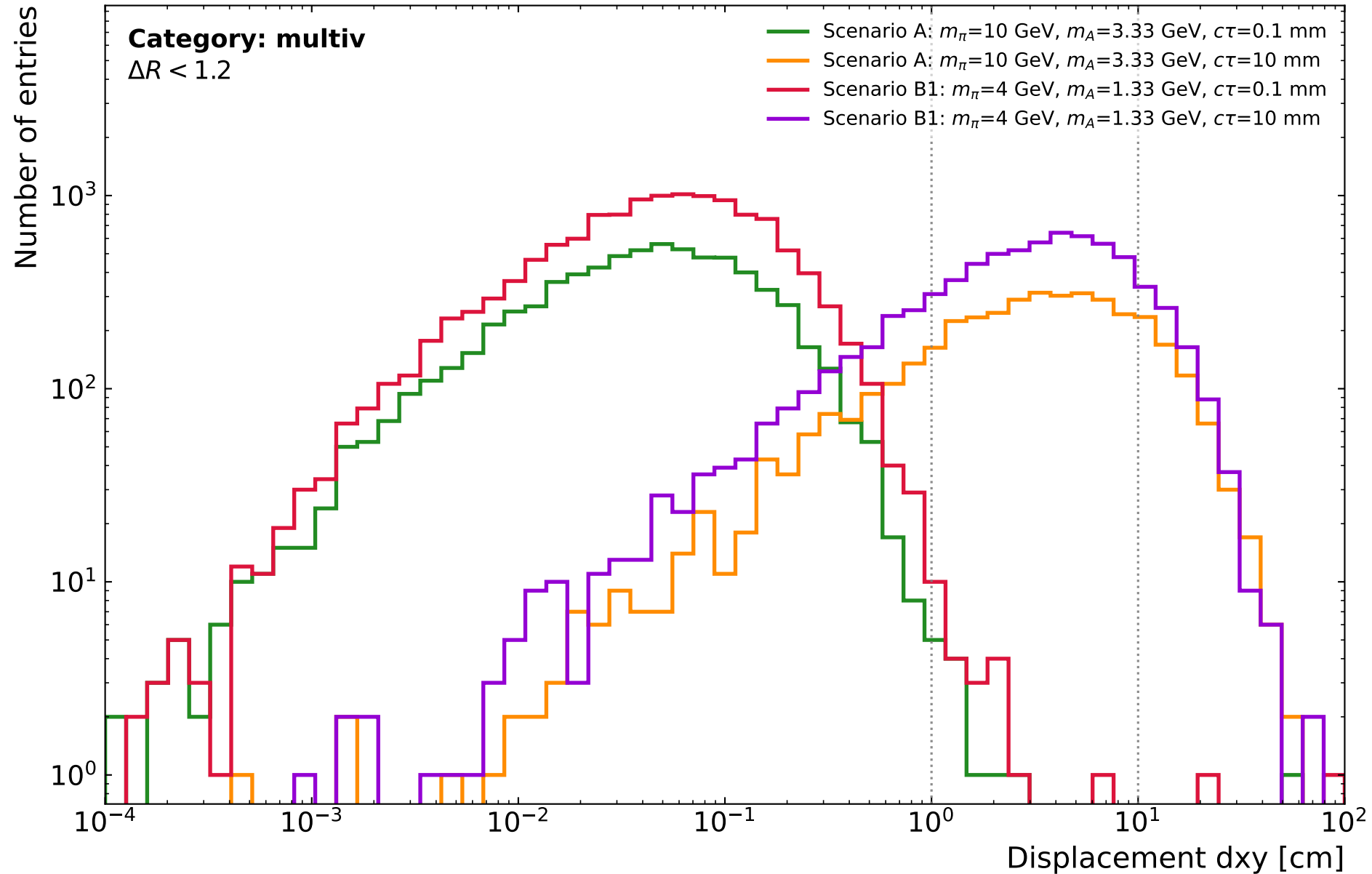


	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	1,629	0	0	1,629
scA $c\tau=10$	0	0	0	0
scB1 $c\tau=0.1$	5,184	0	0	5,184
scB1 $c\tau=10$	121	386	30	537

multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary

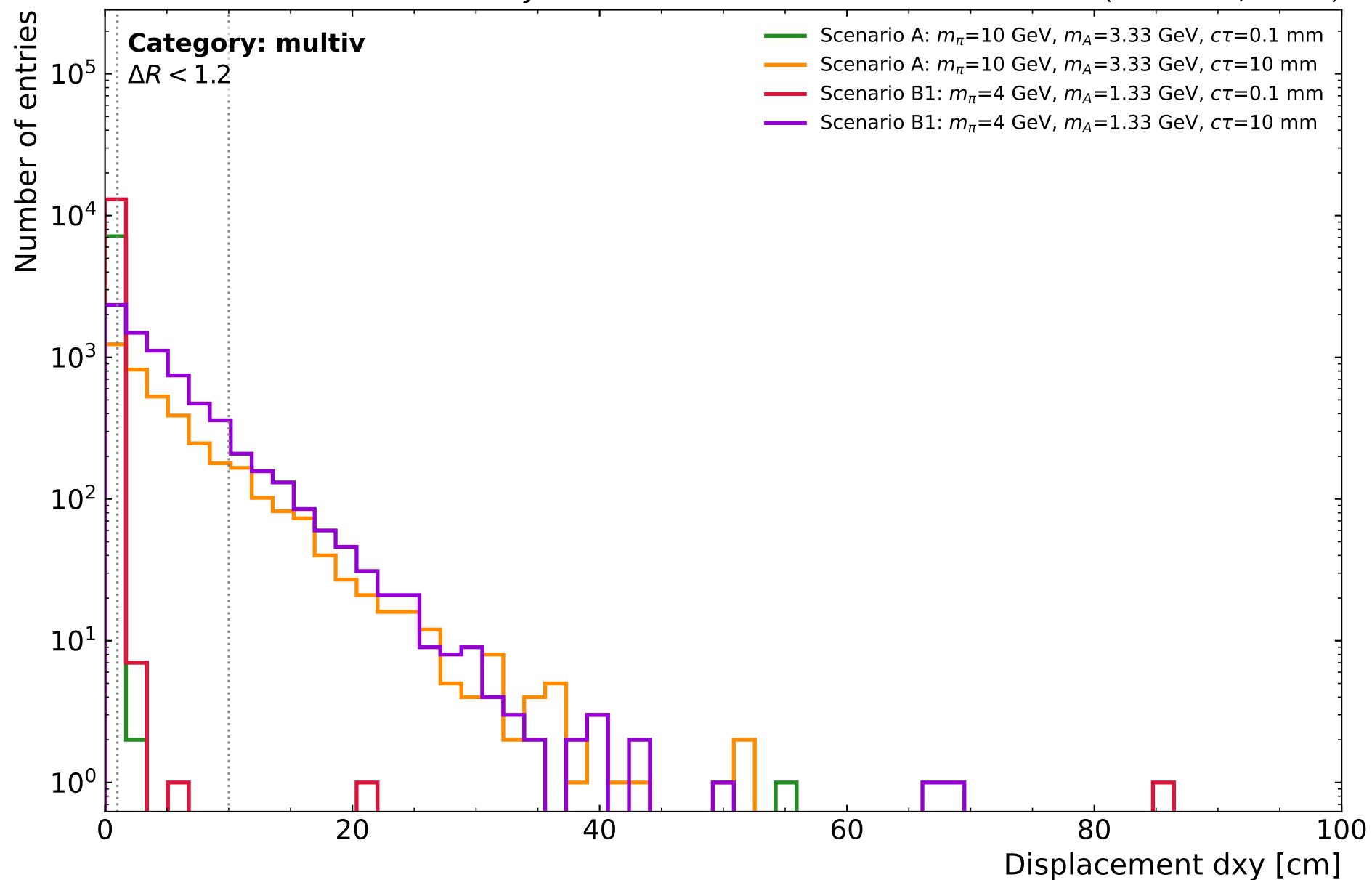
(13.6 TeV, 2024)



multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

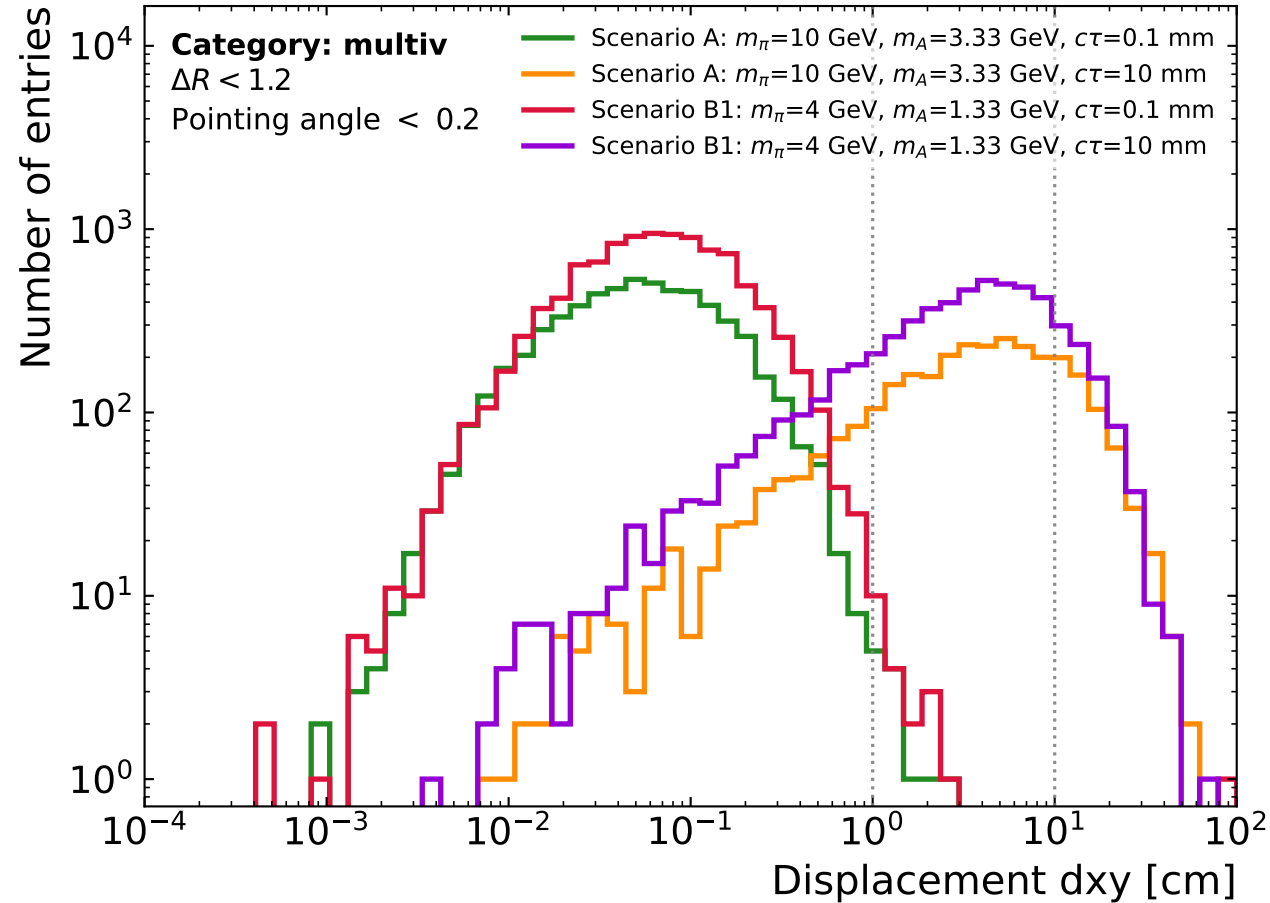
CMS Simulation Preliminary

(13.6 TeV, 2024)



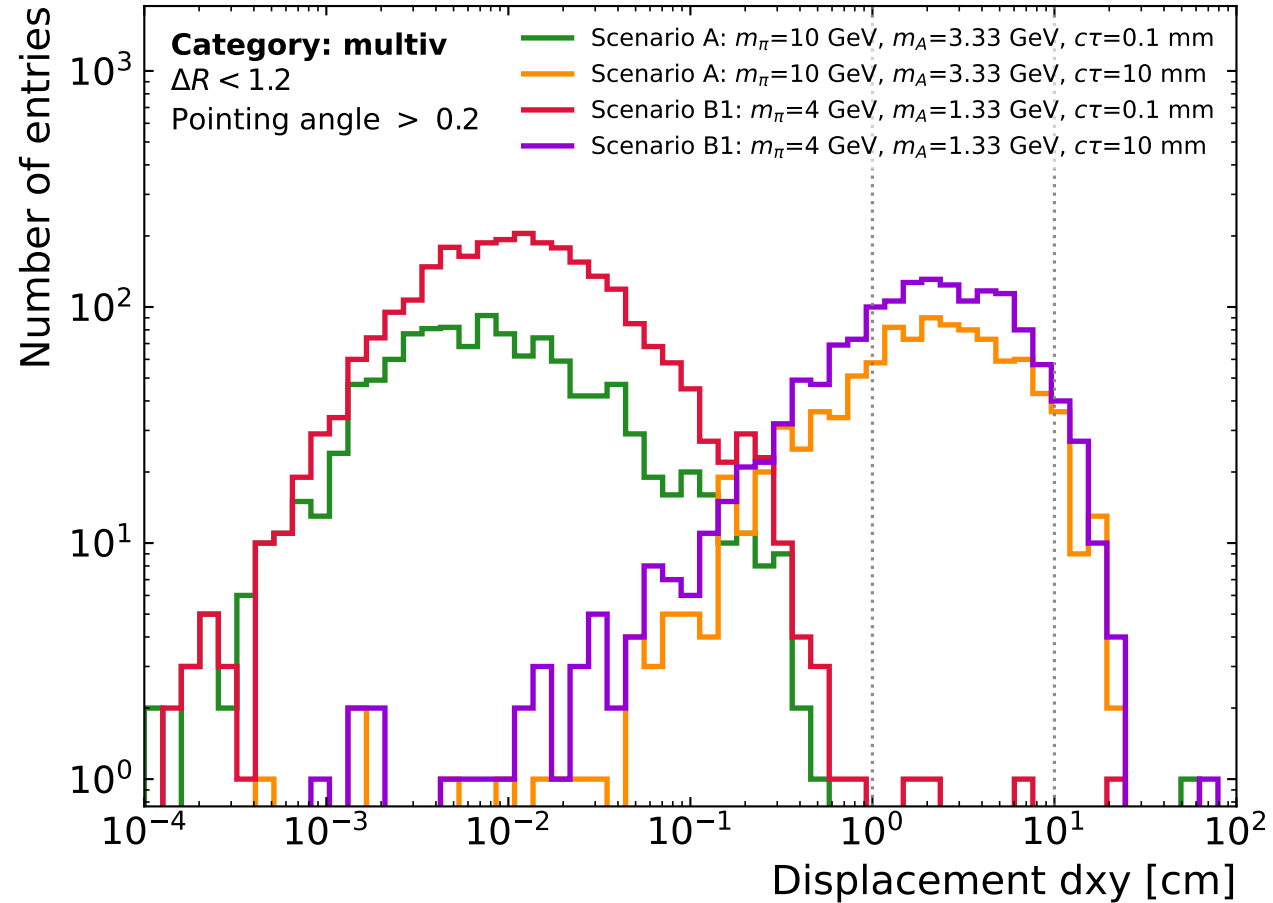
multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
 $\Delta R < 1.2$; dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	5,948	9	0	5,957
scA $c\tau=10$	501	1,922	547	2,970
scB1 $c\tau=0.1$	10,324	18	1	10,343
scB1 $c\tau=10$	1,108	3,924	761	5,793

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
scA $c\tau=0.1$	1,191	0	1	1,192
scA $c\tau=10$	279	685	55	1,019
scB1 $c\tau=0.1$	2,680	3	1	2,684
scB1 $c\tau=10$	422	1,039	72	1,533